

SpaceX Falcon 9 first stage Landing Prediction

Data Science Capstone

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EXECUTIVE SUMMARY



- Data Analysis performed on available data to predict whether a SpaceX Falcon 9 first stage booster will successfully land.
- The cost of a SpaceX launch is much lower than competitors, 62 million USD versus over 165 million USD for competitors.
 - The main reason for this cost difference is SpaceX ability to reuse first stage boosters of the Falcon 9 rocket.
 - If a SpaceX Falcon 9 rocket booster fails to successfully land, the launch cost increases significantly.
- Several factors were examined using multiple models to determine elements that affect the potential success of a Falcon 9 booster landing.

INTRODUCTION



SpaceX is a company founded in 2002 with the aim to make space travel cheaper and more accessible, initially focused on travel to Mars.

SpaceX has since developed a commercial business around using its own rockets to deliver payloads to space.

SpaceX has attempted to reduce the cost of launching rockets by making more components reusable, with a large savings coming from being able to reuse the first stage booster on the Falcon 9 rocket.

SpaceX successfully reusing first stage boosters is factor in being able to deliver payloads to space at a significantly lower cost than other space agencies.

Data collection and data wrangling methodology

- Data collected using SpaceX API
- Scrape data from webpages using BeautifulSoup
- Collect and make sure the data is in the correct format from and SpaceX API using pandas, numpy and python.

```
static_json_url='https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/API_call_spacex_api.json'
response=requests.get(static_json_url)
data = pd.json_normalize(response.json())
data.head(5)

#filter the dataframe
data_falcon9 = pandas_launch_dict[pandas_launch_dict['BoosterVersion']!='Falcon 1']
data_falcon9.loc[:, 'FlightNumber'] = list(range(1, data_falcon9.shape[0]+1))
payload_mean = data_falcon9[
data_falcon9['PayloadMass'] E231 missing whitespace after ' ' (pycodestyle) e=True)
```

```
] static_url = "https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922"

headers = {
    "User-Agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) "
    "AppleWebKit/537.36 (KHTML, like Gecko) "
    "Chrome/91.0.4472.124 Safari/537.36"
}

response = requests.get(static_url, headers=headers)
soup = BeautifulSoup(response.content, "html.parser")
html_tables = soup.find_all('table')
first_launch_table = html_tables[2]
launch_df=pd.DataFrame(launch_dict)
df= pd.DataFrame({ key:pd.Series(value) for key, value in launch_dict.items() })
```

EDA and interactive visual analytics methodology

- Utilize matplotlib and pandas to display scatterplots and bar charts demonstrating success rate relationship to payload mass, orbit, launch site and flight number.

Predictive analysis methodology

Predictive analysis is performed by

1. Accumulating and processing the data
2. Standardizing the data
3. Split the data into testing and training sets
4. Create 4 GridSearchCV objects to model the data
5. Testing the accuracy of each model



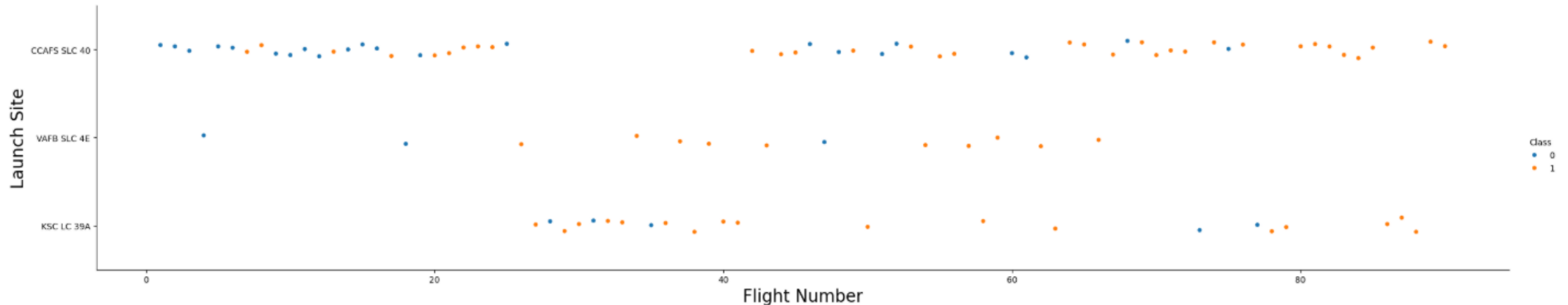
EDA with visualization results (1 of 6)

Exploratory Data Analysis and Feature Engineering

TASK 1: Visualize the relationship between Flight Number and Launch Site

Use the function `catplot` to plot `FlightNumber` vs `LaunchSite`, set the parameter `x` parameter to `FlightNumber`, set the `y` to `Launch Site` and set the parameter `hue` to `'class'`

```
In [5]: # Plot a scatter point chart with x axis to be Flight Number and y axis to be the launch site, and hue to be the class value
sns.catplot(y="LaunchSite", x="FlightNumber", hue="Class", data=df, aspect = 5)
plt.xlabel("Flight Number",fontsize=20)
plt.ylabel("Launch Site",fontsize=20)
plt.show()
```

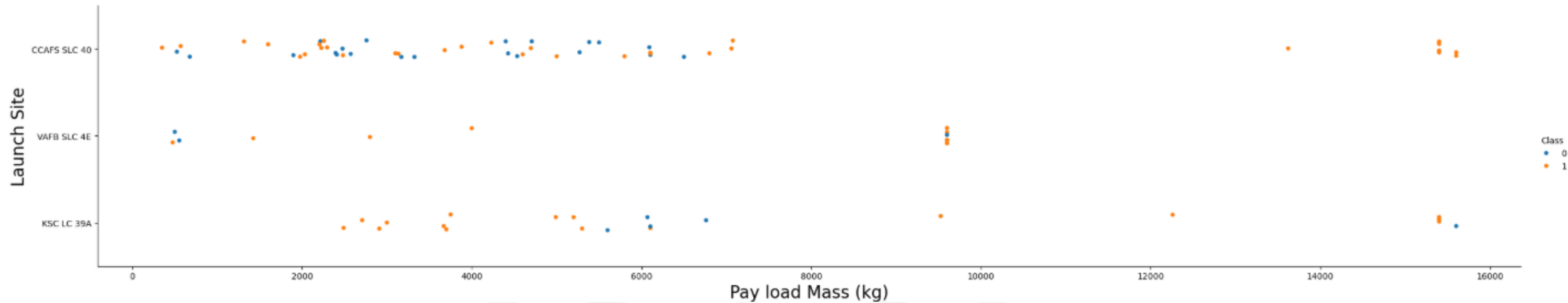


EDA with visualization results slides (2 of 6)

TASK 2: Visualize the relationship between Payload Mass and Launch Site

We also want to observe if there is any relationship between launch sites and their payload mass.

```
In [6]: # Plot a scatter point chart with x axis to be Pay Load Mass (kg) and y axis to be the launch site, and hue to be the class value
sns.catplot(y="LaunchSite", x="PayloadMass", hue="Class", data=df, aspect = 5)
plt.xlabel("Pay load Mass (kg)",fontSize=20)
plt.ylabel("Launch Site",fontSize=20)
plt.show()
```



EDA with visualization results slides (3 of 6)

TASK 3: Visualize the relationship between success rate of each orbit type

Next, we want to visually check if there are any relationship between success rate and orbit type.

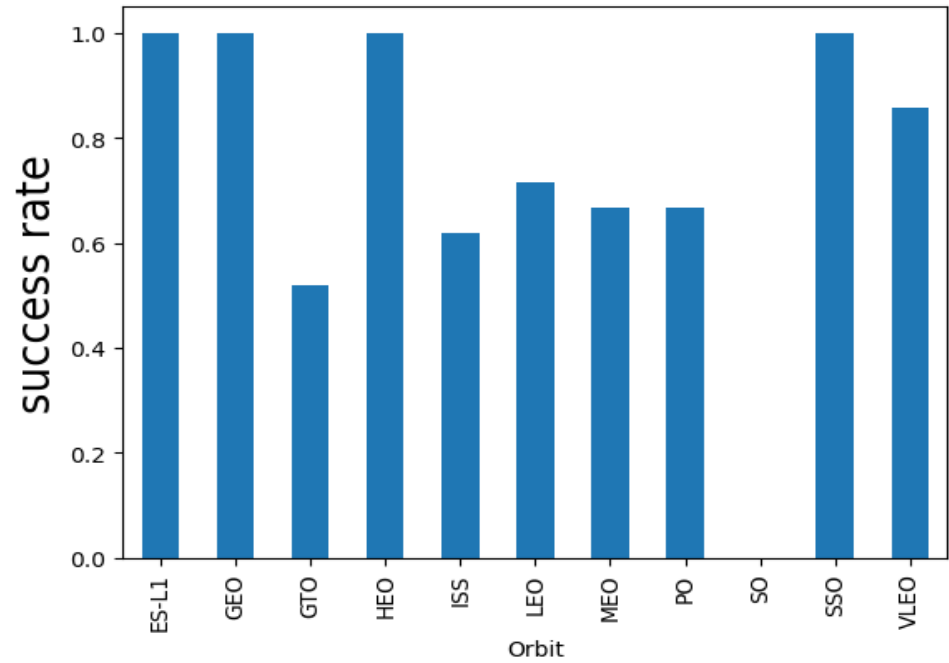
Let's create a `bar` chart for the success rate of each orbit

```
In [37]: df.head(1)
```

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial	Longitude	Latitude	Class
0	1	2010-06-04	Falcon 9	6104.959412	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0003	-80.577366	28.561857	0

```
In [7]: # HINT use groupby method on Orbit column and get the mean of Class column
df_group = df.groupby(['Orbit'])['Class'].mean().plot(kind='bar')
plt.ylabel("success rate",fontsize=20)
```

Out[7]: Text(0, 0.5, 'success rate')

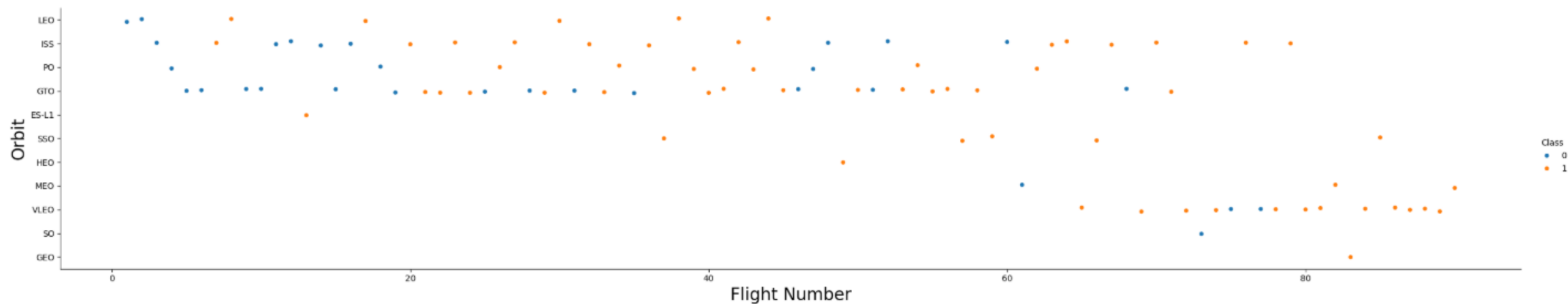


EDA with visualization results slides (4 of 6)

TASK 4: Visualize the relationship between FlightNumber and Orbit type

For each orbit, we want to see if there is any relationship between FlightNumber and Orbit type.

```
In [8]: # Plot a scatter point chart with x axis to be FlightNumber and y axis to be the Orbit, and hue to be the class value
sns.catplot(y="Orbit", x="FlightNumber", hue="Class", data=df, aspect = 5)
plt.xlabel("Flight Number",fontsize=20)
plt.ylabel("Orbit",fontsize=20)
plt.show()
```

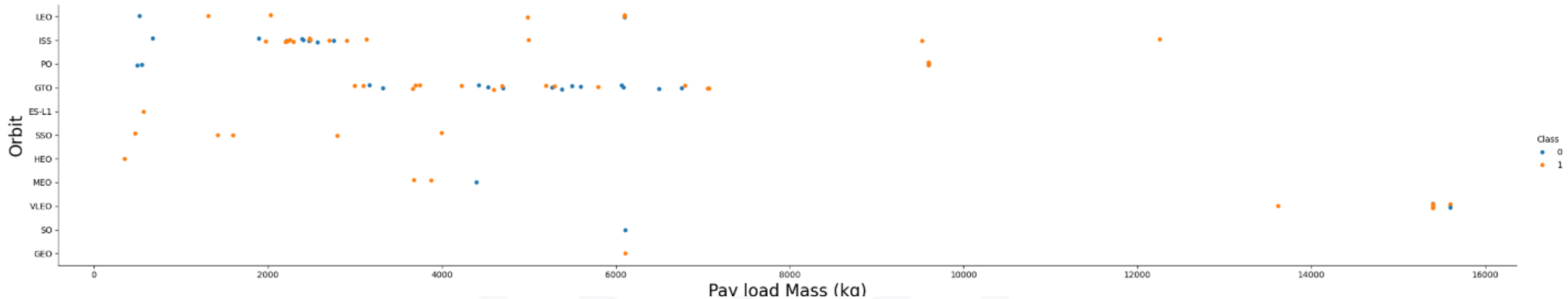


EDA with visualization results slides (5 of 6)

TASK 5: Visualize the relationship between Payload Mass and Orbit type

Similarly, we can plot the Payload Mass vs. Orbit scatter point charts to reveal the relationship between Payload Mass and Orbit type

```
In [9]: # Plot a scatter point chart with x axis to be Payload Mass and y axis to be the Orbit, and hue to be the class value
sns.catplot(y="Orbit", x="PayloadMass", hue="Class", data=df, aspect = 5)
plt.xlabel("Pay load Mass (kg)",fontsize=20)
plt.ylabel("Orbit",fontsize=20)
plt.show()
```



EDA with visualization results slides (6 of 6)

Out[10]:

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial	Longitude	Latitude	Class
0	1	2010	Falcon 9	6104.959412	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0003	-80.577366	28.561857	0
1	2	2012	Falcon 9	525.000000	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0005	-80.577366	28.561857	0
2	3	2013	Falcon 9	677.000000	ISS	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0007	-80.577366	28.561857	0
3	4	2013	Falcon 9	500.000000	PO	VAFB SLC 4E	False Ocean	1	False	False	False	NaN	1.0	0	B1003	-120.610829	34.632093	0
4	5	2013	Falcon 9	3170.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1004	-80.577366	28.561857	0

In [11]:

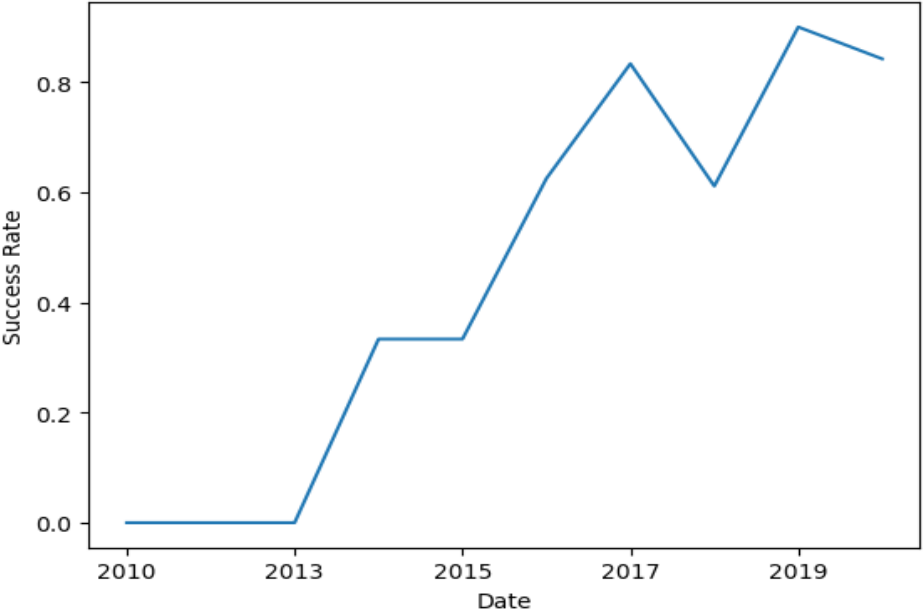
```
success_rate = list(df['Class'])
```

In [12]:

```
df_group_dates = df.groupby(['Date'])['Class'].mean().plot(kind='line')
plt.ylabel('Success Rate')
```

Out[12]:

```
Text(0, 0.5, 'Success Rate')
```



EDA with SQL results (1 of 5)

The dataset includes a record for each payload carried during a SpaceX mission into outer space.
The following SQL queries lead to a table showing types of landing outcomes with number of each

Task 1

Display the names of the unique launch sites in the space mission

```
In [18]: %sql SELECT DISTINCT Launch_Site from SPACEXTABLE;
```

* sqlite:///my_data1.db
Done.

Out[18]:

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
In [21]: %sql SELECT * from SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

* sqlite:///my_data1.db
Done.

Out[21]:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

EDA with SQL results (2 of 5)

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
In [41]: %sql SELECT SUM(sub.PMKG) FROM (SELECT PAYLOAD_MASS_KG_ AS PMKG FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)') as sub;
```

```
* sqlite:///my_data1.db  
Done.
```

```
Out[41]: SUM(sub.PMKG)  
45596
```

```
In [102]: %sql SELECT SUM(PAYLOAD_MASS_KG_ ) AS Total_PayloadMass FROM SPACEXTABLE WHERE Customer LIKE 'NASA (CRS)'
```

```
* sqlite:///my_data1.db  
Done.
```

```
Out[102]: Total_PayloadMass  
45596
```

Task 4

Display average payload mass carried by booster version F9 v1.1

```
In [57]: %sql SELECT AVG(sub.F9_Booster_Weights) FROM (SELECT PAYLOAD_MASS_KG_ AS F9_Booster_Weights FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1') as sub;
```

```
* sqlite:///my_data1.db  
Done.
```

```
Out[57]: AVG(sub.F9_Booster_Weights)  
2928.4
```

EDA with SQL results (3 of 5)

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
In [59]: %sql SELECT MIN(Date) FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';
```

```
* sqlite:///my_data1.db  
Done.
```

```
Out[59]: MIN(Date)
```

```
2015-12-22
```

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
In [65]: %sql SELECT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS_KG_ < 6000 AND PAYLOAD_MASS_KG_ > 4000;
```

```
* sqlite:///my_data1.db  
Done.
```

```
Out[65]: Booster_Version
```

```
F9 FT B1022
```

```
F9 FT B1026
```

```
F9 FT B1021.2
```

```
F9 FT B1031.2
```

Task 7

List the total number of successful and failure mission outcomes

```
In [104...]: %sql SELECT COUNT(Mission_Outcome) AS SuccessOutcome FROM SPACEXTABLE WHERE Mission_Outcome = 'Success';
```

```
* sqlite:///my_data1.db  
Done.
```

```
Out[104...]: SuccessOutcome
```

EDA with SQL results (4 of 5)

Task 8

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
In [75]: %sql SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE AS mplkg;
```

```
* sqlite:///my_data1.db  
Done.
```

```
Out[75]: MAX(PAYLOAD_MASS__KG_)  
15600
```

```
In [109... %sql SELECT DISTINCT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ IN(SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);
```

```
* sqlite:///my_data1.db  
Done.
```

```
Out[109... Booster_Version  
F9 B5 B1048.4  
F9 B5 B1049.4  
F9 B5 B1051.3  
F9 B5 B1056.4  
F9 B5 B1048.5  
F9 B5 B1051.4  
F9 B5 B1049.5  
F9 B5 B1060.2  
F9 B5 B1058.3  
F9 B5 B1051.6  
F9 B5 B1060.3  
F9 B5 B1049.7
```



EDA with SQL results (5 of 5)

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

In [113]:

```
%sql SELECT Booster_Version, Launch_Site, Landing_Outcome FROM SPACEXTABLE WHERE Landing_Outcome LIKE 'Failure (drone ship)' AND Date BETWEEN '2015-01-01' AND '2015-12-31'
```

* sqlite:///my_data1.db
Done.

Out[113]:

Booster_Version	Launch_Site	Landing_Outcome
F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

In [97]:

```
%sql SELECT Landing_Outcome, SUM(nested_value) AS Num_Of_Landings FROM (SELECT Landing_Outcome, SUM(Landing_Outcome) AS nested_value FROM SPACEXTABLE WHERE Date > '2010-06-04' AND Date < '2017-03-20')
```

* sqlite:///my_data1.db
Done.

Out[97]:

Landing_Outcome	Num_Of_Landings
Failure (parachute)	0.0

In [119]:

```
%sql SELECT Landing_Outcome, COUNT(Landing_Outcome) FROM SPACEXTABLE WHERE Date > '2010-06-04' AND Date < '2017-03-20' GROUP BY Landing_Outcome ORDER BY COUNT(Landing_Outcome) DESC;
```

* sqlite:///my_data1.db
Done.

Out[119]:

Landing_Outcome	COUNT(Landing_Outcome)
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2

Interactive map results using Folium (1 of 4)

- Using Folium:
- Mark all launch sites on a map
- Mark the success/failed launches for each site on the map
- Calculate the distances between a launch site to its proximities

Interactive map results using Folium (2 of 4)

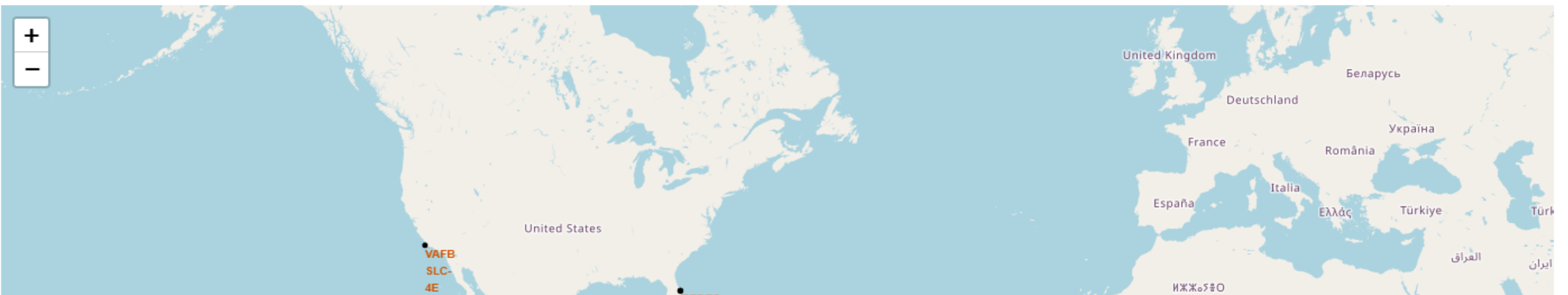
Mark all launch sites on a map

```
# Include the map
site_map = folium.Map(location=nasa_coordinate, zoom_start=5)
# For each launch site, add a Circle object based on its coordinate (Lat, Long) values. In addition, add Launch site name as a popup label
circle1 = folium.Circle(location1, radius=1000, color='#000000', fill=True).add_child(folium.Popup(popup1))
circle2 = folium.Circle(location2, radius=1000, color='#000000', fill=True).add_child(folium.Popup(popup2))
circle3 = folium.Circle(location3, radius=1000, color='#000000', fill=True).add_child(folium.Popup(popup3))
circle4 = folium.Circle(location4, radius=1000, color='#000000', fill=True).add_child(folium.Popup(popup4))

marker1 = folium.map.Marker(location1, icon=DivIcon(icon_size=(20,20),icon_anchor=(0,0), html='<div style="font-size: 12; color:#d35400;"><b>%s</b></div>' % 'CCAFS LC-40', ))
marker2 = folium.map.Marker(location2, icon=DivIcon(icon_size=(20,20),icon_anchor=(0,0), html='<div style="font-size: 12; color:#d35400;"><b>%s</b></div>' % 'CCAFS SLC-40', ))
marker3 = folium.map.Marker(location3, icon=DivIcon(icon_size=(20,20),icon_anchor=(0,0), html='<div style="font-size: 12; color:#d35400;"><b>%s</b></div>' % 'KSC LC-39A', ))
marker4 = folium.map.Marker(location4, icon=DivIcon(icon_size=(20,20),icon_anchor=(0,0), html='<div style="font-size: 12; color:#d35400;"><b>%s</b></div>' % 'VAFB SLC-4E', ))

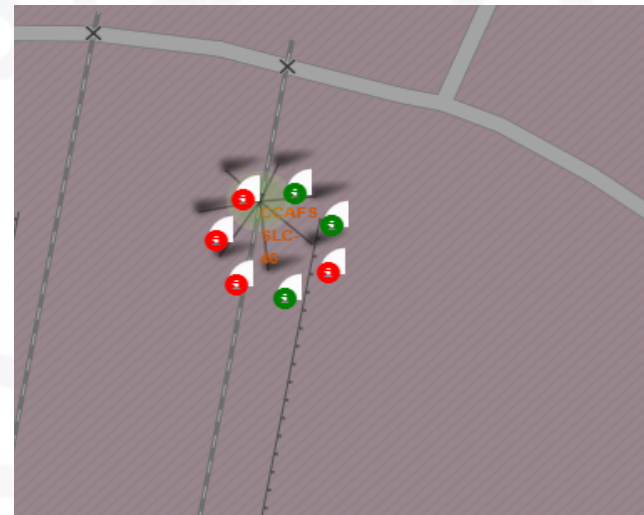
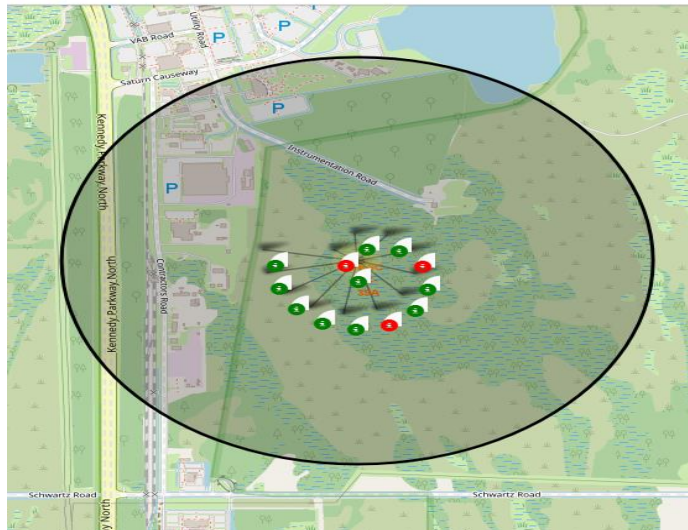
site_map.add_child(circle1)
site_map.add_child(circle2)
site_map.add_child(circle3)
site_map.add_child(circle4)

site_map.add_child(marker1)
site_map.add_child(marker2)
site_map.add_child(marker3)
site_map.add_child(marker4)
```



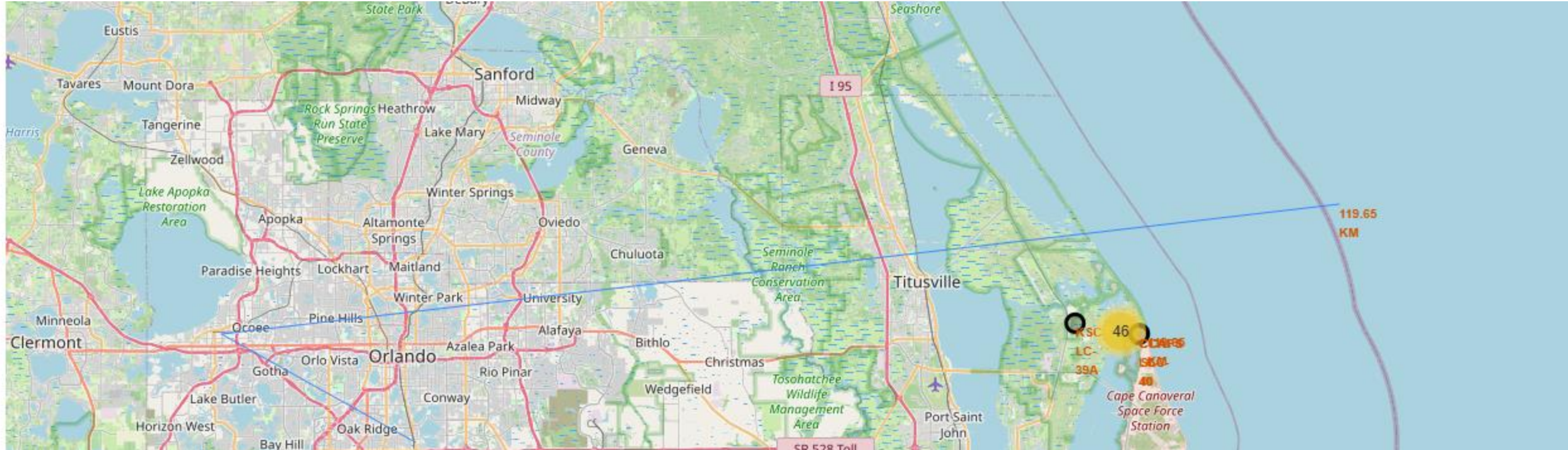
Interactive map results using Folium (3 of 4)

Mark the success/failed launches for each site on the map



Interactive map results using Folium (4 of 4)

Calculate the distances between a launch site to its proximities



Plotly Dash dashboard results (1 of 6)

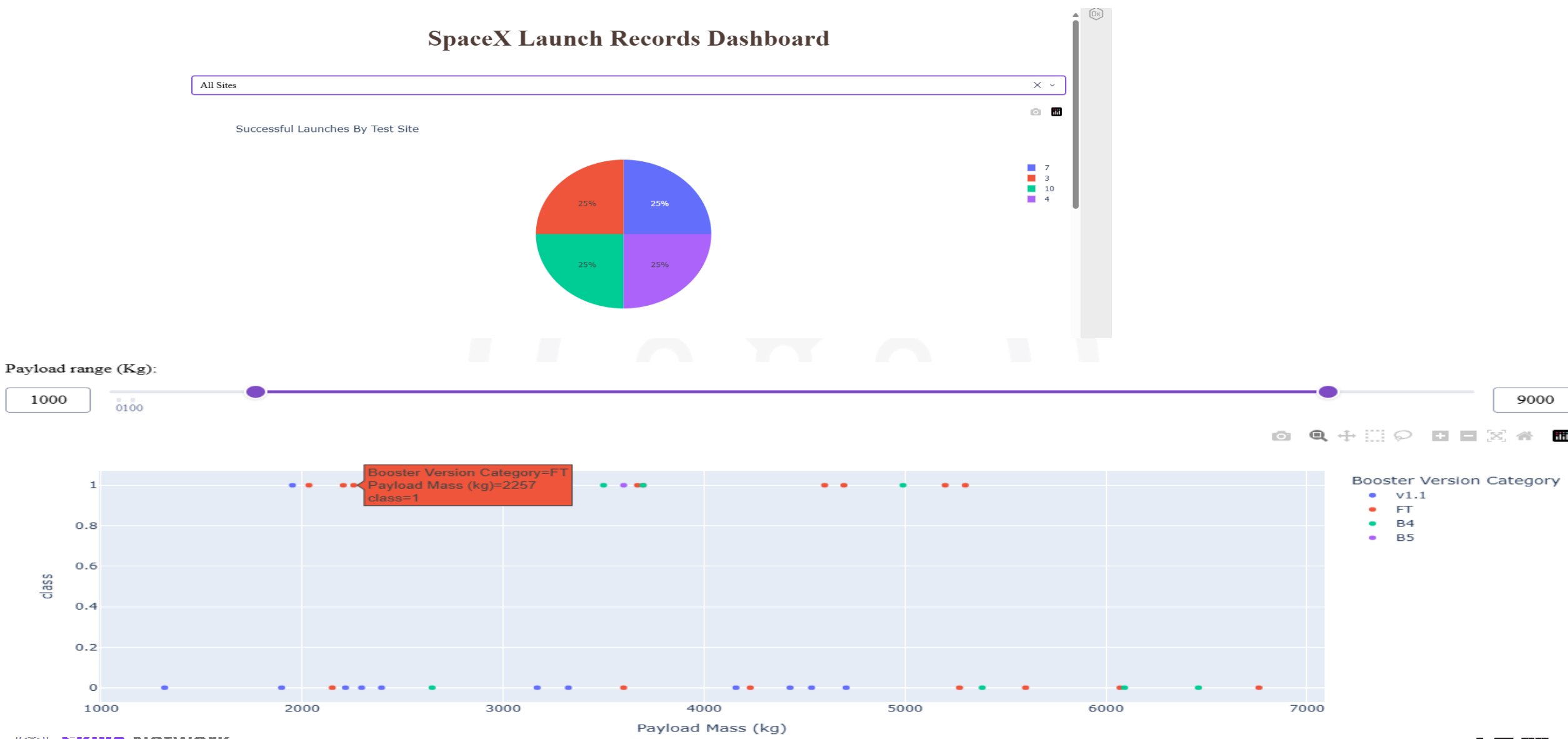


Dash application for users to perform interactive visual analytics on SpaceX launch data in real-time to answer the following questions:

- Which site has the largest successful launches?
- Which site has the highest launch success rate?
- Which payload range(s) has the highest launch success rate?
- Which payload range(s) has the lowest launch success rate?



Plotly Dash dashboard results slides (2 of 6)

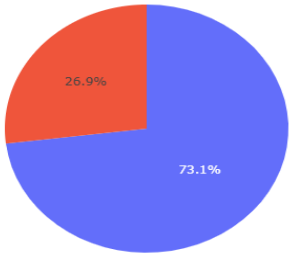


Plotly Dash dashboard results slides (3 of 6)

SpaceX Launch Records Dashboard

CCAFS LC-40

Successful Launches From CCAFS LC-40



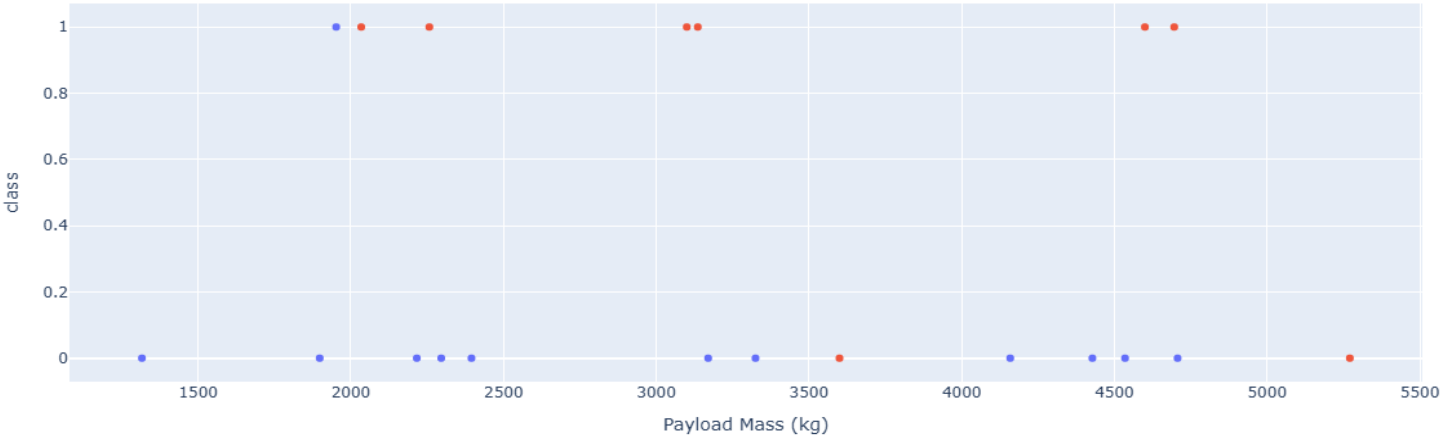
0
1

Payload range (Kg):



Booster Version Category

v1.1
FT

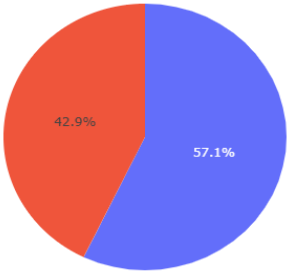


Plotly Dash dashboard results slides (4 of 6)

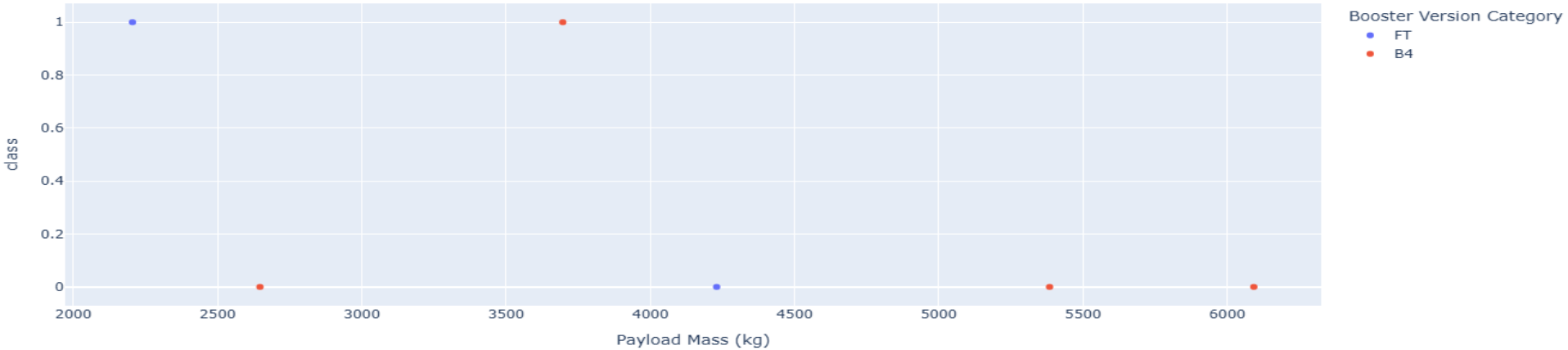
Dash application for users to perform interactive visual analytics on SpaceX launch data in real-time

CCAFS SLC-40

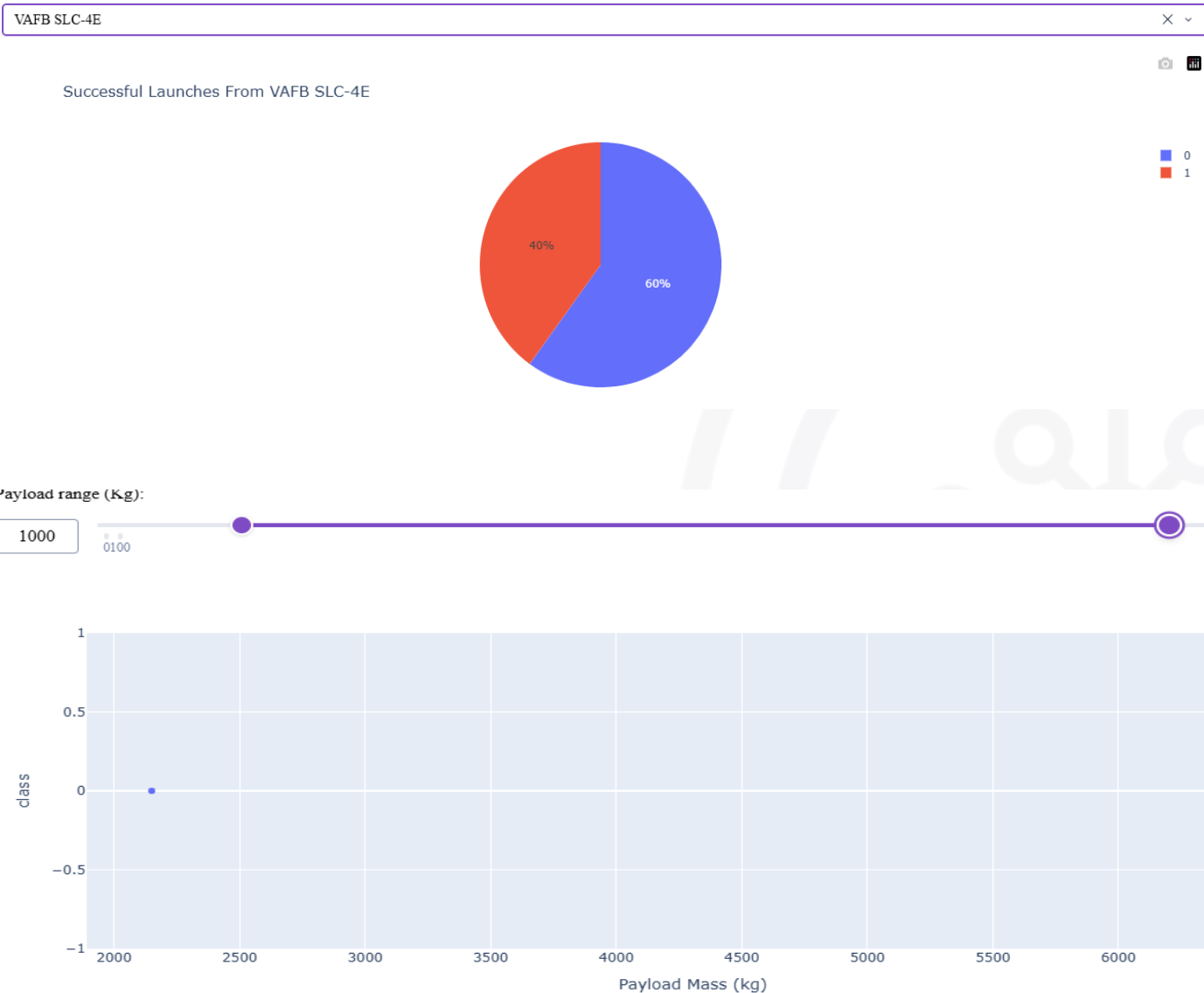
Successful Launches From CCAFS SLC-40



Payload range (Kg):



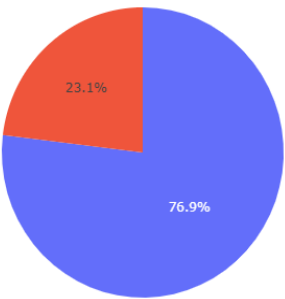
Plotly Dash dashboard results slides (5 of 6)



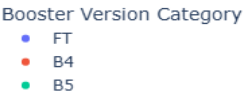
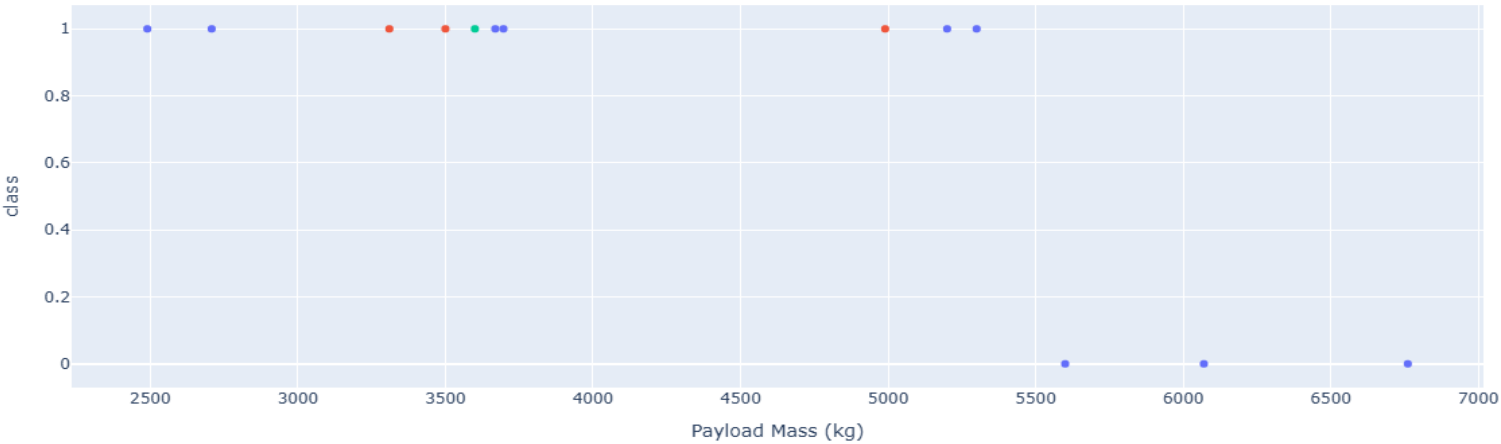
Plotly Dash dashboard results slides (6 of 6)

KSC LC-39A ✕ ▾

Successful Launches From KSC LC-39A



Payload range (Kg):



Predictive analysis (classification) results

```
In [68]: Report = pd.DataFrame({'Method' : ['Test Data Accuracy']})

knn_accuracy=knn_cv.score(X_test, Y_test)
Decision_tree_accuracy=tree_cv.score(X_test, Y_test)
SVM_accuracy=svm_cv.score(X_test, Y_test)
Logistic_Regression=logreg_cv.score(X_test, Y_test)

Report['Logistic_Reg'] = [Logistic_Regression]
Report['SVM'] = [SVM_accuracy]
Report['Decision Tree'] = [Decision_tree_accuracy]
Report['KNN'] = [knn_accuracy]

Report.transpose()
```

```
Out[68]:
```

	0
Method	Test Data Accuracy
Logistic_Reg	0.833333
SVM	0.833333
Decision Tree	0.833333
KNN	0.833333

*All the methods perform equally on the test data: i.e. They all have the same accuracy of *0.833333* on the test Data*



Conclusion

There is correlation between frequency of launches and the success of the launches.

The success rate dramatically improved from zero.

The predictive models all performed nearly the same.

appendix/ github link

Github Link Coursera Data Science Capstone:

https://github.com/bjbci/data_science_capstone/tree/main

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